B30801 Managerial Accounting
Autumn 2025

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CVP/Contribution Margin:
FinePrint
ZPhone

CHICAGO BOOTH

The University of Chicago Booth School of Business

Fineprint: Baseline scenario.

		Vol. = 150K	per unit
		$	$
Manufacturing costs:			
	DM -- variable	6,000	4
	DL -- variable	1,500	1
	DL -- fixed	3,000	
	Maft overhead -- var	1,500	1
	Maft overhead -- fixed	3,375	
Other costs:			
	Sales -- variable	1,500	1
	Sales -- fixed	1,875	
	Corporate -- fixed	3,750	
Note: One unit = 100 brochures			
	Volume = 1,500 units		

Revenues	25,500	17
Variable costs:		
DM	6,000	4
DL	1,500	1
Maft overhead	1,500	1
Sales	1,500	1
Total variable costs	10,500	7
Contribution margin	15,000	10
Fixed costs:		
DL	3,000	
Maft overhead	3,375	
Sales	1,875	
Corporate	3,750	
	12,000	
Pre-tax profit	3,000	

Q1a. Consider the Special Order.

Abbie offers to take 250 units of our output for a price of $10. In addition, we would avoid the variable sales cost of $1/unit.

Q1a.

Because we're at capacity, if we accept the order, we reduce sales to current customers by 250 units/month and replace with sales to Abbie:

Δ Revenue =	(250 x $10) – (250 x $17)	-1,750
Δ Costs =	-(250 x $1)	250
Δ Bottom-line		-1,500

Q1a.

Because we're at capacity, if we accept the order, we reduce sales to current customers by 250 units/month and replace with sales to Abbie:

Δ Revenue =	(250 x \$10) – (250 x \$17)	-1,750
Δ Costs =	-(250 x \$1)	250
Δ Bottom-line		-1,500

Alternatively, compare unit contribution margin of special order of \$4 (\$10 – (\$4 + \$1 +\$1)) to that of our normal orders of \$10. Because capacity is constrained, we should allocate to the product with the highest per unit contribution margin, which is the existing sales.

So **reject special order**.

Key idea: When capacity is constrained, allocate capacity to the product with the highest contribution margin per unit of the constrained resource.

Q1b.

Here, we have available capacity and so should accept any orders that generate positive contribution margin. In this case, the contribution margin is $4/unit (see above) and so we accept the special order.

Incremental pre-tax profit: 250 x $4 = +$1,000.

Q1b.

Here, we have available capacity and so should accept any orders that generate positive contribution margin. In this case, the contribution margin is \$4/unit (see above) and so we accept the special order.

Incremental pre-tax profit: 250 x \$4 = +\$1,000.

Key idea: When we have available capacity, accept all new orders that generate a positive contribution (i.e., assuming they have no effects on existing sales or operations).

Q2.

The answer here depends on what you assume. There are two possibilities.

First, assume we continue to sell 150K brochures (1,500 units) but outsource production of 30K brochures (300 units) to Bradley. So revenue doesn't change and it's just a matter of comparing costs:

Δ Revenue =		
Δ Costs =	\$8 – (\$4 + \$1 +\$1) = \$2	-600
Δ Bottom-line		-600

Because costs increase by \$600 with no other changes, we reject the proposal (contribution margin falls by \$2/unit).

Q2.

Alternatively (and probably more likely) you might assume that we can sell the new 300 units at the current price of $17. These units have a contribution margin of $8 ($17 - $8 - $1) (we still incur the variable selling cost).

Because the cm > 0, we accept the proposal. The total increment to pre-tax profit is 300 x $8 = +$2,400.

Unit Contribution Margins:

	Model K	Model X
Unit price	$750	$1,000
DL cost @ $18	(27)	(36)
Battery cells @ $10	(10)	(20)
Processor	(45)	(45)
Touch screen	(50)	(75)
Other materials	(150)	(170)
Total VC	(282)	(346)
Unit CM $	$468	$654
DL hours	1.5	2
Battery cells	1	2

Q1.				
	Breakeven point (units):		6,410.26	= FC/cm
	So breakeven is		**6,411**	
	Fixed costs		$3,000,000	
	Because unit CM > 0 should produce max. possible, so 11,000 units.			
	EBIT =		$2,148,000	
	See Fig. 1			

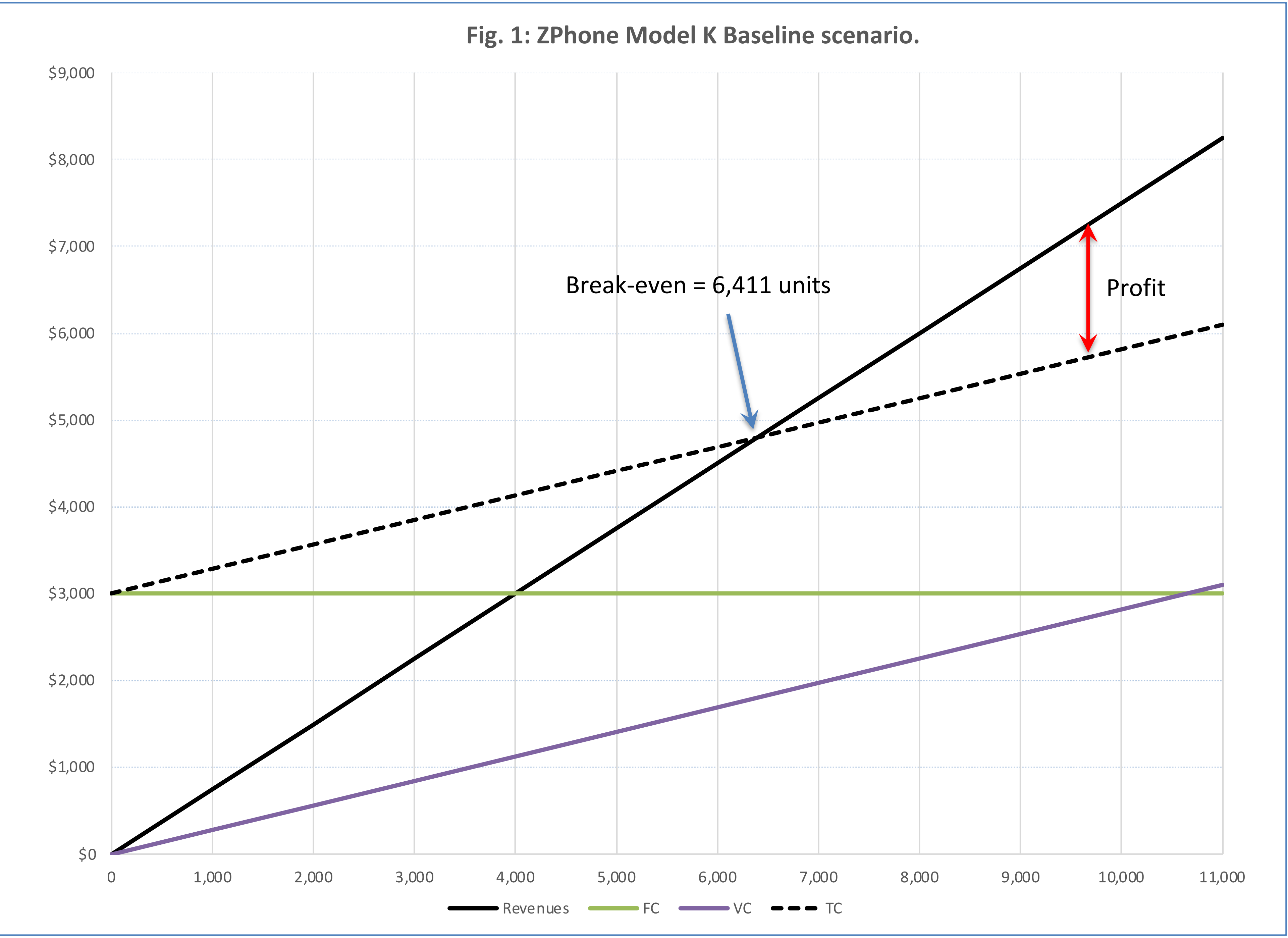


Fig. 1: ZPhone Model K Baseline scenario.
$9,000
$8,000
$7,000
$6,000
$5,000
$4,000
$3,000
$2,000
$1,000
$0
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
9,000
10,000
11,000
Break-even = 6,411 units
Profit
Revenues
FC
VC
TC

Q2.					
		Breakeven point (units):		13,888.89	= FC/cm
		So breakeven is		**13,889**	
		Fixed costs		6,500,000	
		Because unit CM > 0 should produce max. possible, so 20,000 units.			
		EBIT =		$2,860,000	

There are now two break-even points: 6,411 and 13,889 depending on whether the second shift is added.

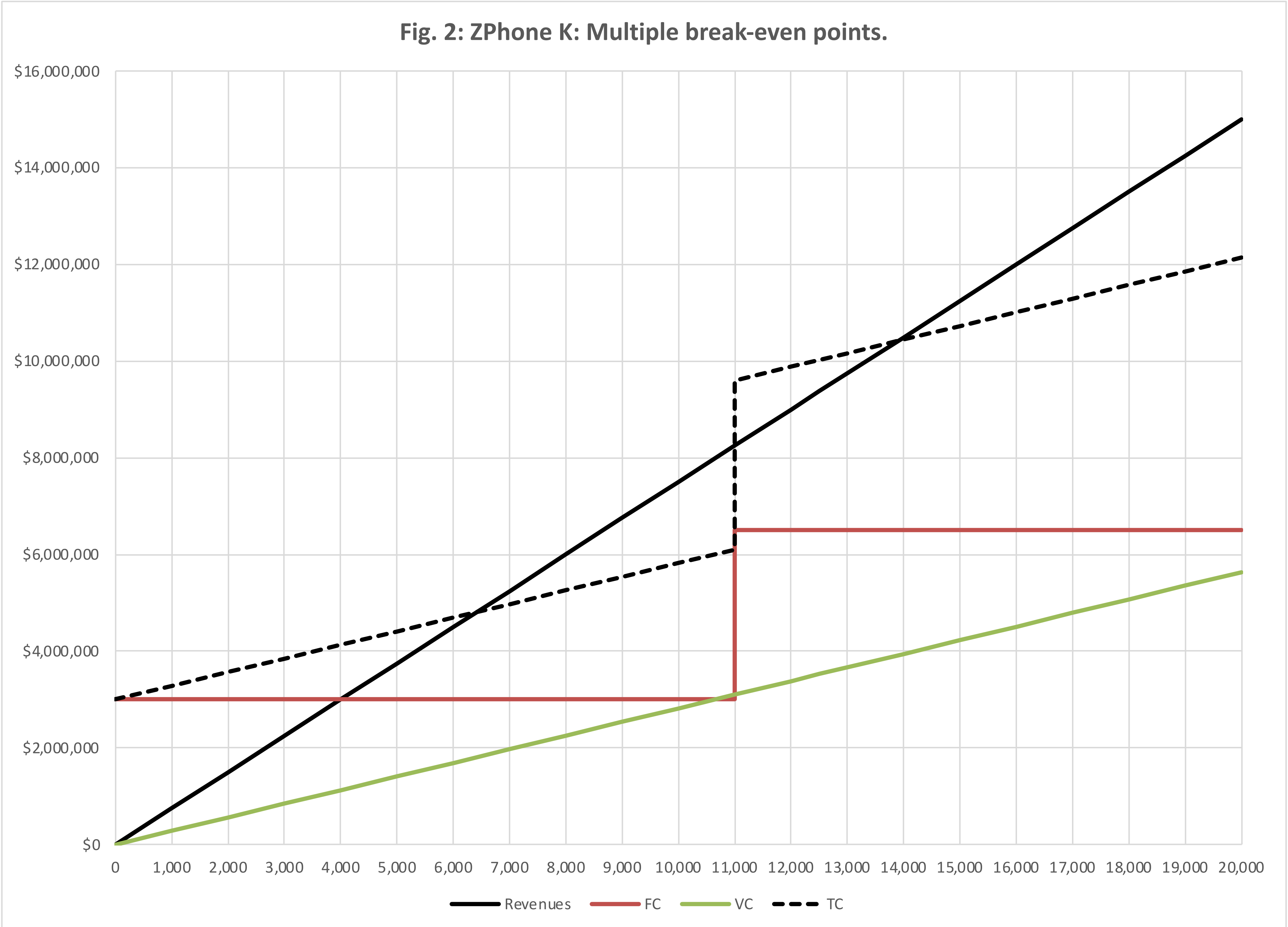


Fig. 2: ZPhone K: Multiple break-even points.
$16,000,000
$14,000,000
$12,000,000
$10,000,000
$8,000,000
$6,000,000
$4,000,000
$2,000,000
$0
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
9,000
10,000
11,000
12,000
13,000
14,000
15,000
16,000
17,000
18,000
19,000
20,000
Revenues
FC
VC
TC

Q3.

The answer here is straightforward: because Model X has higher unit CM than Model K (and that $CM > 0$) we should produce as many units of Model X as possible.

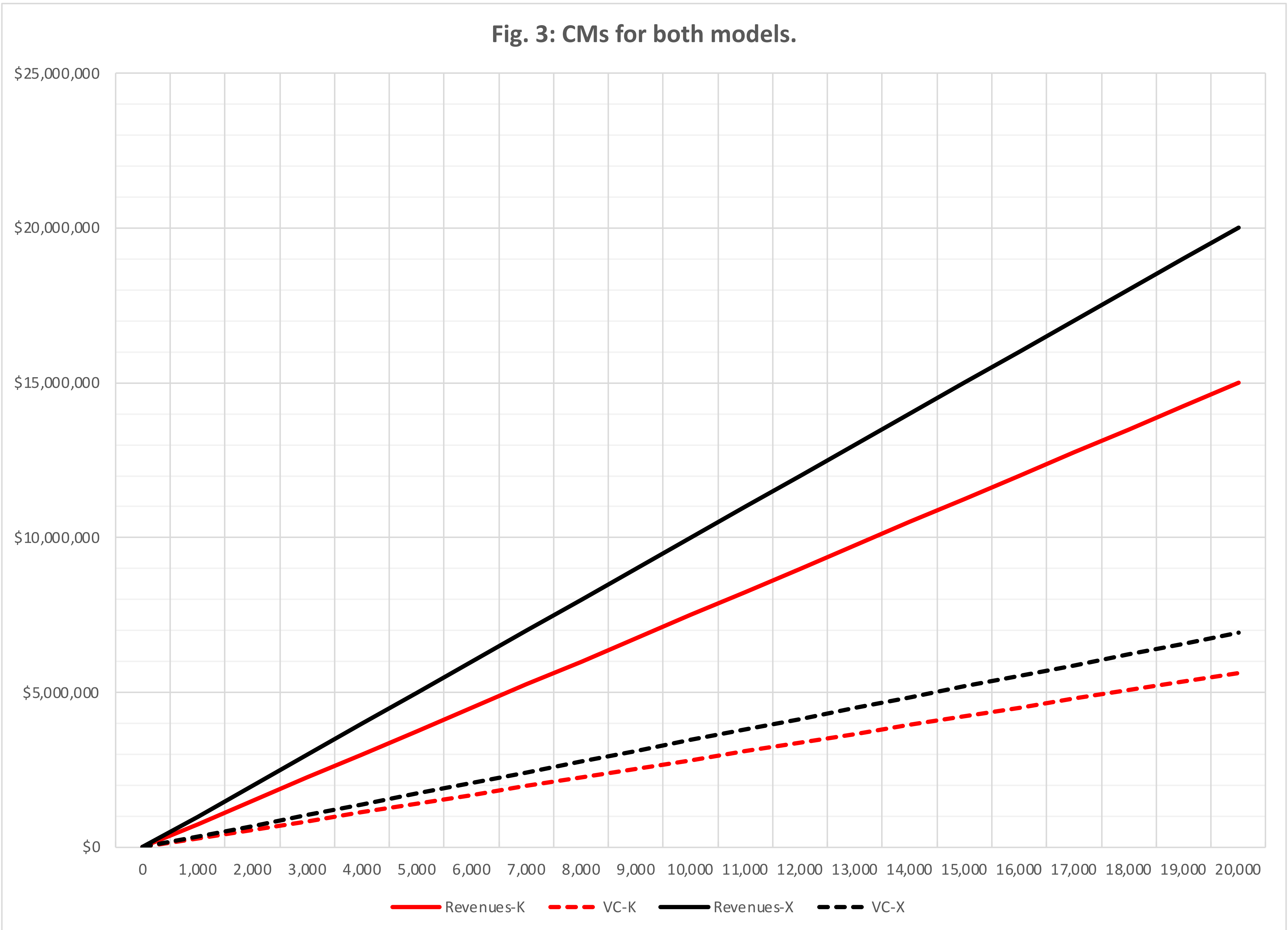


Fig. 3: CMs for both models.
$25,000,000
$20,000,000
$15,000,000
$10,000,000
$5,000,000
$0
0
1,000
2,000
3,000
4,000
5,000
6,000
7,000
8,000
9,000
10,000
11,000
12,000
13,000
14,000
15,000
16,000
17,000
18,000
19,000
20,000
Revenues-K
VC-K
Revenues-X
VC-X

Q4.

Goal is to maximize total CM$. What happens if we only produce K?

Can produce 10,000 for total CM of $4,680.

What about if instead we only produce X?

Given the constraint, we only produce 5,000 for total CM of $3,270.

So we should **produce 10K units of Model K.**

The general way to approach these problems is to compute the CM per unit of the constrained resource and maximize this. Here, the constraint is battery supply, so we need CM per battery:

Model K	$468
Model X	$327

Which again tells us to produce Model K (i.e., allocate all capacity to Model K).

The general way to approach these problems is to compute the CM per unit of the constrained resource. Here, the constraint is battery supply, so we need CM per battery:

Model K	\$468
Model X	\$327

Which again tells us to produce Model K (i.e., allocate all capacity to Model K).

Another way to understand this is to think about producing an incremental unit of X (which generates \$654 of CM):
- to do this requires that we reduce production of K by 2 units, reducing CM by \$936, so there's a net reduction of \$282.

Q5.

Under the contribution margin approach, total CM would increase by 500 x $468, and so by $234,000.

This tells us something about how much the constraint is costing the business (as well as how much we'd be prepared to pay for an additional battery). Of course, we are also assuming that fixed costs don't change (we are still within the initial relevant range).

Q6.

If, in addition, we can only obtain 8,000 processors per month, we need to set this up as follows:

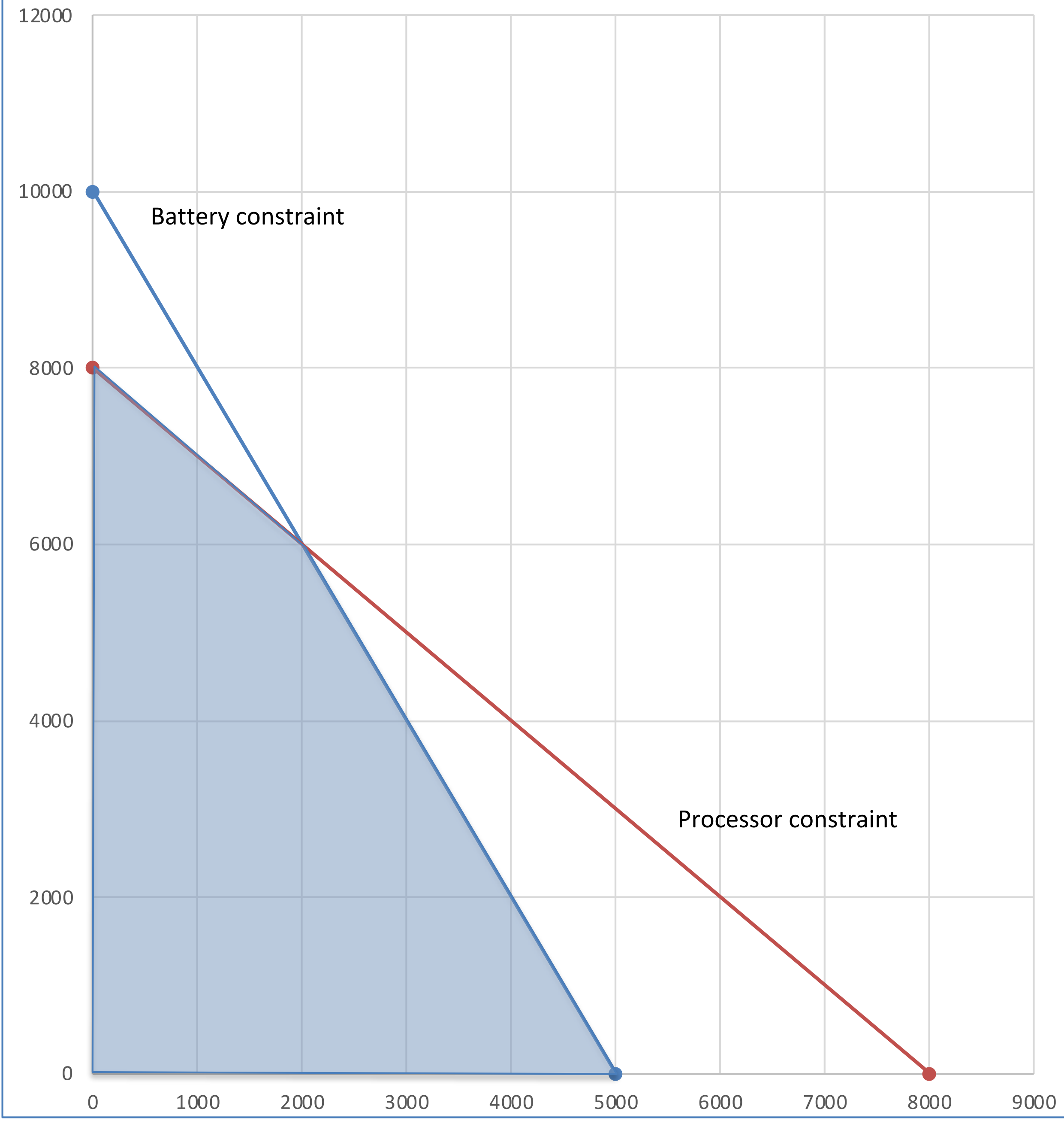
Max $\$468K + \$654X$

subject to:

$$
\begin{aligned}
2X + K &\leq 10{,}000 \\
X + K &\leq 8{,}000 \\
X, K &\geq 0
\end{aligned}
$$

This is a linear programming problem that can be solved graphically (since there are two variables)

Fig. 4: ZPhone with two constraints



Here, the vertical axis is the number of units of K, the horizontal axis is the number of units of X.

The shaded area is the feasible set (all available possibilities).

We need to evaluate each intersection point of the lines that form the shaded area:

Units K	Units X	Total CM
8,000	0	$3,744,000
0	5,000	$3,270,000
6,000	2,000	$4,116,000

The solution is to produce 6,000 units of K and 2,000 units of X, which maximizes overall CM.